

Carola Bonamico

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Education

- Politecnico di Torino**, MS in Computer Engineering – Software Sep 2024 – Present
- Grade average: 27.8/30
 - Coursework: Data Science and Database Technology, Software Engineering I/II, Web Applications I, LLM For Software Engineering, Human Computer Interaction, Computer Architectures, Information Systems, System and Device Programming
- University of Pavia**, BS in Bioengineering Sep 2020 – Apr 2024
- Thesis: “Literature analysis regarding the impact of sleep quality on glycemic patterns in patients with type 1 diabetes”
 - Coursework: Biomedical Data Processing, Models of Biological Systems
- I.I.S. Cellini – Liceo Linguistico Alberti**, Linguistic Diploma Sep 2015 – Jun 2020
- Languages: English, French, German

Experience

- Master’s Thesis Student**, ETH Zurich – Zurich, CH Mar 2026 – Present
- Conducting the Master’s thesis project at the Integrated Systems Laboratory (IIS), focused on the development of Deep Learning models for silent-speech recognition from electromyographic signals.
- Software Developer**, Squadra Corse PoliTo – Torino, IT Aug 2025 – Nov 2025
- Development of a TUI in Go to send, receive, and inspect CAN messages via SocketCAN interface, optimizing on-track diagnostics and debugging.
- Academic Tutor**, University of Pavia – Pavia, IT Sep 2022 – Oct 2023
- Tutoring activities for small groups of students in the Fundamentals of Computer Science course and targeted individual support on C programming exercises, covering algorithm design and memory management.

Technical Skills

Programming Languages: C, Rust, Go, JavaScript, TypeScript, Python, ARM Assembly, MATLAB, $\LaTeX$

Deep Learning: PyTorch, NumPy, Pandas

Libraries and Frameworks: Node.js, React.js, React Native, TypeORM, Diesel

Databases: Oracle SQL, MongoDB, MySQL, SQLite, PostgreSQL

Tools and Software: Git, GitHub, GitLab, Docker, Jupyter Notebook, LabVIEW

Other: REST APIs, WebSockets, JSON, Repository Pattern, Agile, Scrum, UML, HTML, CSS

Personal Projects

- CAN debug TUI** github.com/squadracorsepolito/can-debug
- CAN bus debugging TUI developed in Go, which implements sending and receiving of CAN messages via SocketCAN interface with individual message control. Supports DBC files for message decoding.
- Autostar Motorsport Web App** github.com/carolabonamico/autostarmotorsport-webapp
- Responsive web application for a motorsport company, built with Node.js and React.js. Implements a JavaScript client-side architecture, GDPR compliance via Iubenda, and modular React components styled with CSS and AOS.

Projects at ETH Zurich

- SilentWear** github.com/carolabonamico/SilentWear
- Fork of SilentWear, a wearable system based on a textile neckband for silent speech recognition via EMG. Extension of the existing CNN with a BiLSTM and a Transformer. Implementation of CTC decoding with beam search for direct EMG-to-text prediction. Integration of a new dataset collected on 7 subjects with a custom protocol, for approximately 38 hours of recording, including isolated words and full sentences in vocalized and silent conditions.
- BioGUI** github.com/carolabonamico/biogui
- Contributed to BioGUI, an open-source real-time biosignal acquisition and labeling platform. Extended the data collection pipeline to support a custom vocabulary protocol. Implemented intra-file alignment between EMG, EEG, and microphone signals within the custom .bio binary format, as well as inter-file alignment across multiple .bio recordings.

Projects at Politecnico di Torino

- Participium** github.com/MauroC0l/Participium
- Full-stack civic reporting platform in TypeScript/JavaScript (Node.js and web frontend), with role-based workflows, geolocation, and photo uploads. Includes Telegram notifications and SonarQube for continuous code quality monitoring and static analysis.
- Rust Chat** github.com/carolabonamico/ruggine-chat
- Real-time group messaging application developed with Rust and React. Supports private chats, groups, JWT authentication, WebSocket communication, and Docker deployment.

HCI MoveMate github.com/carolabonamico/HCI-movemate

Repository of documentation and reports related to the HCI research and UX design process, including results, decision rationales, and evaluation outcomes supporting the MoveMate prototype.

MoveMate github.com/carolabonamico/movemate

High-fidelity mobile app prototype built with React Native (Expo), focused on community event discovery and participation through an end-to-end mobile UX flow.

Geocontrol github.com/carolabonamico/geocontrol

Dockerized software monitoring system for managing geological, meteorological, and environmental sensors. Built in TypeScript following the repository pattern, with TypeORM and Node.js.

Class Assignments Manager github.com/carolabonamico/class-assignments-manager

Web application for managing assignments from teachers to student groups. JavaScript server-side, React.js client-side with React Bootstrap and CSS.

Pac-Man on LandTiger github.com/carolabonamico/pacman-landtiger

Pac-Man video game for the LandTiger LPC1768 board (ARM Cortex-M3) implementing A* algorithm for ghost movement, pause functionality, and game audio. Written in C and ARM Assembly.

Projects at University of Pavia

Clinical Trial Manager github.com/carolabonamico/clinical-trial-manager

Website for managing clinical trials developed in HTML and CSS, supporting communication between doctors, patients, and clinical staff, with monitoring and follow-up of clinical studies.

Pulse Oximeter Data Analysis github.com/carolabonamico/analisi-dati-pulsossimetro

Software developed in LabVIEW for processing data acquired from a pulse oximeter designed and assembled in the laboratory.

National Health Service (SSN) Activity Management Application

github.com/carolabonamico/applicativo-gestione-attivita-SSN

MATLAB application for managing activities of regions, ASL (local health authorities), and hospitals, with statistical analysis on patient glycemic curves using XML data processing.

Certifications

Goethe-Zertifikat B1 (German), Goethe-Institut Italia

Issued Jul 2019

B1 Preliminary (English), Cambridge English

Issued May 2018